
Input contract
(PyTeal/Rust, x)
Semantic description generation
d = gθ(x)
schema matches KB indexing
d
Query embedding
v_query = f_enc(d) ∈ ℝ^d
Cosine-similarity search
sim(v_q,v_i)=
(v_q·v_i)/(‖v_q‖‖v_i‖)
S=argtop distinct-class sim,
|S|=N
top-k idx
Dynamic checklist prompt assembly
full detail if rank_sim(i) ≤ N
prompt = λ(x, checklist(S))
Audit & verdict generation
CoT, then <final> block
ŷ~pθ(·|prompt)
Security report
verdict ŷ, see callout ↓
inference
LLM call
(shared model)
query v_query
Vector KB
K={(c_i,v_i)}, i=1..M
(vuln. snippets + fields)
snippet + vuln. info (top-N classes)
inference
LLM call
(same model)
excerpt of x (PyTeal)
Gtxn[0].rekey_to() ==
Global.zero_address()
Gtxn[1].rekey_to() ==
Global.zero_address()
⚠ missing on Gtxn[0]; checked on Gtxn[1] only
output d (example)
"Gtxn[1] requires rekey_to ==
ZeroAddress; Gtxn[0]: no such check."
this is d, fed right into f_enc
✓ Vulnerability: Rekey-To
Explanation: Gtxn[0] unchecked
Risk: High, full account takeover
Mitigation: check rekey_to on every txn
(parsed from <final> block)
matches excerpt on the left
Deterministic step
LLM inference call
Persistent vector store
inference-time call